

ATT SIS LEARN

Electronics & Robotics Programme

Course Structure, Session Plan & Pricing

Instructor

Bharanidharan Thamizhchelvan

P.Eng · Electronics Design Engineer
10+ years · Medical Device & Embedded
Systems

Programme details

Ages 6–18 · Elementary to High School
London, Ontario · Weekends · Flexible
Small group · Max 5 students per cohort
First session completely FREE

About Attsis Learn

Attsis Learn is a small-group, hands-on electronics and robotics programme for children and teenagers in London, Ontario. Every session is designed around building something real — not watching videos, not worksheets. Students leave each session with a working project they built themselves.

The programme is designed and taught by a licensed Professional Engineer (P.Eng.) with over 10 years of experience in electronics design, embedded systems, and medical device development. The curriculum follows a structured progression from beginner robotics at age 6 all the way to advanced autonomous systems and drone programming for high school students.

At the end of the programme, every student receives a Certificate of Completion signed by the instructor as a P.Eng., along with a personal observation letter documenting the engineering thinking skills demonstrated throughout the course. This letter can support gifted programme applications, school portfolios, or university preparation.

Programme Levels

Level 1 · Junior Robotics · Ages 6–10 · Elementary (Grades 1–5)

Designed for young learners with no prior experience. Students work with LEGO Education Spike Essential kits to build robots that move, sense, and respond to their environment. Block-based coding introduces programming logic in a visual, intuitive way. By Session 6, students give their first voice command to a machine — the first step toward the drone and AI track.

- How robots see, move, and make decisions
- Motors, sensors, and simple machines
- Block-based coding — sequences, loops, and conditionals
- Obstacle avoidance and sensor-triggered responses
- Light, sound, and signal commands
- Voice command introduction (foundation for drone track)
- Final showcase project — presented to parents

- Certificate of Completion + P.Eng. observation letter

Level 2 · Electronics & Arduino · Ages 11–14 · Secondary (Grades 6–9)

Students move from block coding to real circuit building and programming in C. Using Arduino microcontroller kits, students build circuits on breadboards, read data from sensors, and program autonomous robot behaviours. The session on drone flight principles introduces aerodynamics and control systems — laying the foundation for the Advanced level.

- Breadboards, resistors, LEDs — real circuit construction
- Arduino programming in C — variables, functions, logic
- Ultrasonic, temperature, and motion sensors
- Motor control and servo motor logic
- Introduction to drone flight principles and aerodynamics
- Capstone autonomous robot — designed by the student
- Demo Day presentation with certificate

Level 3 · Advanced Systems · Ages 15–18 · High School (Grades 10–12) · Coming Soon

The Advanced level is currently in curriculum development and will open for enrolment in a future cohort. Students who complete Level 2 are automatically priority-listed for Level 3.

- Advanced microcontroller and embedded systems programming
- PCB design fundamentals and soldering
- Autonomous systems — sensor fusion and decision making
- Drone programming and flight control (DJI Tello SDK)
- Capstone real-world engineering project

Session Plan — Level 1 · Junior Robotics

#	Session title	What the student builds / learns	Tag
FREE	Free consultation	Child brings their toy. Instructor observes how they think, explain, and build. Parents present. No curriculum — pure discovery.	FREE
S1	Meet your robot	What makes something a robot? Sort devices. First LEGO build: a car that moves forward and backward.	WOW
S2	Sensors — the robot's eyes	Add a distance sensor. Robot stops before hitting a wall. Child sees cause and effect in real time.	WOW
S3	Coding basics	Block coding on LEGO app. Program the robot to follow a path. Introduction to loops and sequences.	
S4	Making decisions	If/then logic — robot turns left when it detects an obstacle on the right. First conditional	

		thinking.	
S5	Light, sound and signals	Add LED and buzzer. Program light patterns and sound alerts. Robot communicates with the world.	WOW
S6	Voice & commands intro	Simple voice-trigger using tablet. Child gives first spoken command to a machine. Seeds the drone dream.	DRONE
S7	Design your own	Child designs a robot to solve their own problem. Instructor guides only. Parent observes full engineering thinking.	
S8	Final showcase	Child presents robot to parents and explains how it works. Certificate issued. P.Eng. observation letter provided.	CERT

Session Plan — Level 2 · Electronics & Arduino

#	Session title	What the student builds / learns	Tag
FREE	Free consultation	Assessment session. Student explains a circuit or device they know. Instructor calibrates starting level.	FREE
S1	Circuits & components	Breadboard basics. Build a simple LED circuit with resistors. Understand voltage, current, and resistance.	WOW
S2	Arduino first program	Upload first sketch — blink an LED. Edit timing, understand code structure. Hello World of hardware.	
S3	Reading sensors	Connect ultrasonic sensor. Read distance data in Serial Monitor. Build a proximity alarm.	WOW
S4	Motor control	Control a DC motor with Arduino. Add direction switching. Build a remote-controlled robot chassis.	
S5	Servo & precision movement	Program a servo motor for precise angular control. Build a robotic arm segment.	WOW
S6	Drone flight principles	Aerodynamics, lift, thrust, drag. How a drone stays stable. Introduction to flight controllers.	DRONE
S7	Capstone build begins	Student designs their autonomous robot. Instructor guides system architecture. Sensors + motors + logic.	
S8	Demo Day	Student presents capstone robot to parents. Explains design decisions. Certificate + observation letter.	CERT

Pricing & Packages

Option	Level 1 (Ages 6–10)	Level 2 (Ages 11–14)	Sibling Bundle (Both)
Drop-in (per session)	\$44.99 / session	\$53.99 / session	\$80.99 / session (both)
Monthly (4 sessions)	\$161.99 / month	\$197.99 / month	\$296.99 / month (both)
Full bundle (8 sessions)	\$314.99 total	\$377.99 total	\$539.99 total (both)
Savings vs drop-in	Save \$44.93	Save \$53.93	Save \$107.93

Sibling discount: When two children from the same family enroll in different levels simultaneously, the combined session rate drops from \$98.98 to \$80.99 — saving \$107.93 over a full 8-session bundle.

What is included

Item	Included	Notes	Cost
Session instruction (60–75 min)	Yes	P.Eng. instructor, full session	Included
Free trial session (Session 0)	Yes	1 hour, full session, parents present	Free
Trial kit (instructor provides)	Yes	LEGO or Arduino provided for trial only	Free
Session worksheets	Yes	Printed, kept by student	Included
Certificate of completion	Yes	Full bundle only, signed by P.Eng.	Included
P.Eng. observation letter	Yes	Full bundle only — for school portfolios	Included
Session photo update to parents	Yes	WhatsApp after every session	Free
LEGO Spike Essential kit	No	Parent purchases for Level 1	~\$150 CAD
Arduino starter kit	No	Parent purchases for Level 2	~\$60 CAD
Consumables (wires, LEDs, etc.)	Partial	Basic stock provided by instructor	~\$5/session

Payment policy

- Payment method: Interac e-Transfer (Global Attsis)
- Bundle payment: 50% upfront before Session 1, 50% at Session 4
- Drop-in payment: 24 hours before each session
- Cancellation: 48 hours notice required — session credited, not refunded

- Missed session: 1 makeup session included per full bundle
- Receipt issued by email after each payment

Equipment Guide for Parents

Students need their own kit starting from Session 1. The free trial session uses equipment provided by the instructor — no purchase needed before the trial.

After the trial, the instructor will send you the exact Amazon.ca link for your child's kit. Please allow 2–3 days for delivery before Session 1.

Level	Kit	Approx. cost	Notes
Level 1 (Ages 6–10)	LEGO Education Spike Essential	~\$150 CAD	Amazon.ca link sent after trial
Level 2 (Ages 11–14)	Arduino Starter Kit (Elegoo or official)	~\$60 CAD	Amazon.ca link sent after trial

Safety & Compliance

Attsis Learn takes the safety of every student seriously. The following measures are in place before any child attends a session.

- **Instructor has applied for a Vulnerable Sector police background check — standard for all individuals working with children in Ontario. Certificate available on request.**Vulnerable Sector Check (VSC):
- **Parents are welcome and encouraged to stay for any session, especially the free trial. No child is left unsupervised.**Parent presence:
- **Completed before Session 1 begins.**Parent waiver & emergency contact form:
- **Present in the teaching space at all sessions.**First aid kit:
- **Parents are invited to inspect the teaching space before committing to the programme.**Venue inspection:
- **Private tutoring and instruction in Ontario does not require a city childcare licence when parents are present or nearby. Verified with London city bylaws.**Home occupation:
- **Instructor liability coverage in place. Details available on request.**Insurance:

Next Steps

- 1 Book the free 1-hour trial session — bring your child and their favourite robot or electronic toy if they have one.
- 2 Parents observe the full session. The instructor will assess your child's level and enthusiasm.

3	After the trial, the instructor shares a personalised recommendation — which level, which schedule, which kit to order.
4	If you wish to proceed, confirm the bundle or drop-in option, complete the parent waiver, and make the first payment by e-Transfer.
5	Amazon.ca kit link sent within 24 hours. Kit arrives 2–3 days before Session 1 begins.

Ready to book your free trial?

Contact us on WhatsApp · +1 (226) 972-7885

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